

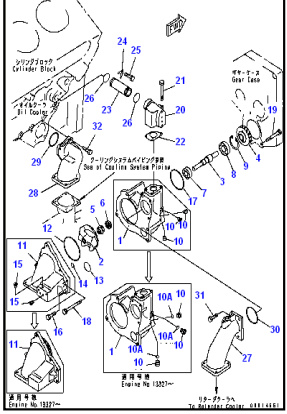
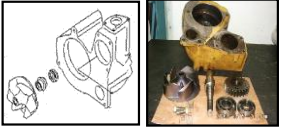
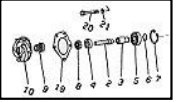
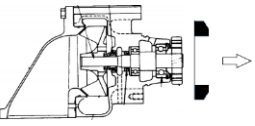
ASSEMBLY CHECK SHEET

WATER PUMP

6D170&12V140 SERIES

WO NUMBER		PART NAME	WATER PUMP	DOCUMENT NO.	RC/QR04/EG/04/02
ENGINE MODEL		PART NUMBER		PUBLIC DATE	25/03/2010
ENGINE S/N		START DATE / TIME		REVISION NO.	2
MACHINE MODEL		FINISH DATE / TIME		PAGE	

ASSEMBLY BY		All dimension in millimetres
		All torques in Kilogram metres

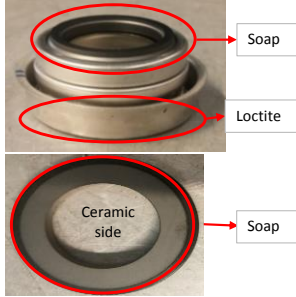
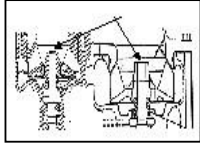
NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA/GL	
1	Pump Body		Pump body must be free from crack, dent and rush Clean All Parts throughly and make sure that all burns have been remove. Always check the contact of the following parts before assembly. See checking contact between parts. Marking with paint maker, for all of bolts and nuts after tightening by torque wrench	See maintenance standard for standard dimension			
2	Housing		Housing must be free from crack, dents and rust All threads are in good condition and clean Make sure that face (clearance area) is in good and clean				
3	Other Parts		Recheck condition of bearings, oil seal, shaft, impeller, water seal, gear, snap ring, bolts and washers should be free from crack, rust, dents, scratch and pitting	Refer and follow PSN			There are many PSN for the Water pump 
4	Insertion of Oil Seal		Oil seal seat must be clean and dry (free from oil, fuel or water) Coat the out side of oil seal with Red Liquid gasket. Insert oil seal using special tools Coat the seal lip with "grease" sufficiently	Use oil seal installer LG 1207 G2-LI			
5	Insertion Bearing to Shaft		Insert small bearing and big bearing to shaft using special tools Make sure that bearings center to shaft when assembly For 12V140 Series Before inserting the small bearing and the big bearing, make sure that the shaft is new shaft with model P/n.6219-61-1230 (shaft end stamping reading "S5")	Use special Tools and Press by Hyd.Jack			 Make sure that bearings center to the shaft when assembly
6	Shaft Assembly		Coat the shaft (seal contact) by sufficient grease Bearings and bearing seat must be clean Insert shaft ass'y to the body and make sure that shaft assy is centering to body CAUTION Recheck and make sure that oil seal isn't hit or damage when inserting the shaft Check shaft from smooth rotation.	Use special Tools (Shaft Installer)			Snap Ring Position  Big Snap ring (at water pump) has different face, please insert chamfer face to outside. 

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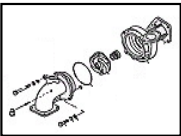
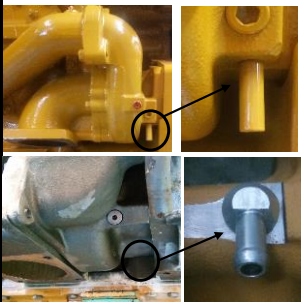
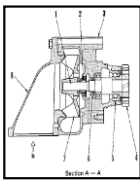
NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA/GL	
7	Insertion of Drive Gear		Shaft and gear contact must be clean Coat the drive gear hole with engine oil Install Spacer. Insert drive gear to the shaft until contact to inner race bearing or stopper Standard of interference between Gear and Shaft Act. Interference : mm	Drive Gear Installer 0.02 ~ 0.06 mm			When inserting the drive gear Please arrest the shaft at the End of the shaft  
8	Insertion of Water Seal		Make sure that the condition of water seal is free from scratch, crack, pitting, dents, corrosive, worn, and etc. Here is critical point, be careful Coat the outer side of water seal with adhesive Insert water seal to housing with special tool. Coat Ceramic surface of water seal with "toilet soap"	Follow and refer to the PSN LOCTITE 609 (green loctite) TOILET SOAP			
9	Insertion of Impeller		Insert water seal using special tools Insert the seal ring to the groove at impeller and make sure that it is fit in the groove Coat the hole of the impeller with small quantity of Engine Oil be carefull when insert the Impeller For 6D170 Series When inserting the impeller , adjust impeller from end of shaft using special tool (spacer) to assure Standard of Spacer Thickness Act. Spacer thickness = mm Clearance between impeller and connector / housing Standard Clearance Act. Clearance = mm For 12V140 Series Before inserting the impeller, make sure that impeller is new impeller model with P/n.6219-61-1470 (change the hole for the shaft) Act. Clearance = mm Insert the impeller to the shaft until contact to the end of the shaft.	SAE 15W-40 1.2 ~ 1.7 mm 0.23 ~ 0.66 mm Standard Clearance : 0.25 ~ 1.02 mm			    

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NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
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10	Insertion of Housing Hole Indicator		O-ring groove must be free from crack, scratch, pitting, dents and rust Check O-ring from scratch Insert O-ring and check from twist and out of it's groove Insert Housing/connection and tighten mounting bolts carefully and properly Act. Torque M8 = Kgm Act. Torque M10 = Kgm Act. Torque M12 = Kgm M8 = 3.2 + 0.3 kgm M10 = 6.7 + 0.7 kgm M12 = 11.5 + 1.0 kgm Install extension tube to hole indicator				  
11	General Inspection		√ : Check - Check Shaft for smooth rotation ☐ OK ☐ - Recheck position of snap ring ☐ OK ☐ - Check loose bolts from mistake ☐ OK ☐ - Check all parts from improper installation ☐ OK ☐ - Check and keep completion parts ☐ OK ☐				 1. Pump shaft 2. Water seal 3. Pump body 4. Water pump drive gear (No. of teeth: 25) 5. Ball bearing 6. Oil seal 7. Impeller 8. Pump cover a. From thermostat b. From radiator (water inlet) c. To oil cooler (water outlet) 
12	End play water pump		Measure End play of water pump shaft before delivery to Main line	Standard End play 0			

MECHANIC ASSEMBLY COMMENT

ASSEMBLY BY	INSPECTED & APPROVED BY
MECHANIC (MECH)	QUALITY ASSURANCE (QA)
	PROD. SPV/GL